

Grounding the Loop on Both Sides: Interaction as a Third Test-Time Compute Axis, and Why Its Gains Are Invisible Without Grounded Evaluation

Bojie Li
Pine AI
boj@19pine.ai

Abstract

Test-time compute is dominated by two scaling axes that both operate inside a model’s own token space: *reasoning* (longer chains of thought) and *sampling* (best-of- N). Both are bounded, by the data-processing inequality (DPI), by the information already in the weights and prompt. A third axis—*interaction* with an environment that returns grounded feedback—can exceed that bound by importing information from outside the model. We argue that a single variable, **grounding**, governs this axis, and that it must hold on *both* sides of the loop: grounded *feedback* drives improvement past the reasoning/sampling ceiling, and grounded *evaluation* is required even to *measure* the improvement. We formalize the first claim as a DPI bound over a Type 0–3 feedback taxonomy, and we demonstrate the second empirically: on rendered visual artifacts the default vision–language-model (VLM) judge is structurally blind to the geometric defects the loop fixes—it rates 14 of 15 broken academic figures “perfect”—and a VLM *reviewer* even makes layouts worse. Replacing it with a deterministic DOM-geometry instrument (overlap, clipping, overflow, and box-group misalignment, measured exactly from the rendered layout) reveals large, significant defect reductions under grounded feedback across **four** visual modalities measured identically: academic figures –74%, dense slides –73%, responsive web pages –47%, and SVG/CSS animations –40% (all paired sign-test $p < 0.002$; all bootstrap 95% CIs exclude zero). On execution-grounded code the harness lifts a frontier model from 66.7% to a strict 100% pass-rate, and an in-modality control isolates the *execution* of tests—not the extra reviewing pass—as the cause; this replicates across three model families and a held-out suite, and we extend the grounded-geometry harness to a second model family (Gemini 3.1 Pro) scored by the same model-free instrument. At a fixed 20K-token budget, reasoning-only and best-of- N saturate (73.3%/86.7%) while every interaction strategy reaches $\geq 97.8\%$, and the proposer–reviewer harness does so at the lowest token cost with zero seed variance. Finally, the harness’s behavior partially distills into an 8B student. Interaction scaling is real, distinct, and predictable—but only visible when both the feedback and the metric are grounded.

1 Introduction

Scaling test-time compute has become the dominant lever for improving large models on artifact-producing tasks—code, web pages, slides, figures, animations, research reports. The literature has formalized two axes, and both operate *inside the model’s own token space*. *Reasoning scaling*

[2, 6, 8, 9] spends more tokens thinking before committing to an answer. *Sampling scaling* [1, 7] draws more independent attempts and selects one. In both, every additional token is a function of the same frozen weights and the same fixed prompt: more compute reorganizes information already in the system but imports none.

A third axis breaks this closure: *interaction* with an external environment that returns **grounded** feedback—a test result, a rendered layout, a search verdict. Such feedback is not a function of the weights; it is an observation of the artifact’s real behavior, and it can carry information the model never had. This axis is well established empirically [3–5, 10], but accounts of *why* and *when* it should beat reasoning- or sampling-only scaling remain informal, and—we argue—half-stated.

Grounding is the variable, on both sides of the loop. Our thesis is that one variable governs interaction scaling: **grounding**, the property that a signal is computed from the artifact’s actual behavior rather than from a model’s opinion of it. And grounding must hold on *both* sides of the proposer–reviewer loop (Figure 1):

1. **Grounded feedback** (the reviewer signal that drives revision): run the code and read `pytest`; render the HTML and measure the real bounding boxes; issue the query and read the result. This is the side prior work emphasizes, and a data-processing-inequality argument (Section 2) shows it is exactly what lets interaction exceed the ceiling that bounds reasoning and sampling.
2. **Grounded evaluation** (the metric that *measures* the gain): the same observation, used as the score. This side is almost always overlooked, and getting it wrong silently hides the effect.

The second point is not a detail. The default evaluator for visual artifacts is a vision–language model (VLM) reading a screenshot, and it is *structurally* blind to the defects that matter: content overflowing the canvas is cropped off-frame before the image reaches the judge, and small overlaps fall below its acuity. On dense academic figures this VLM judge rates 14 of 15 single-shot renders “perfect” when a deterministic check finds only 3 are actually clean (Section 6). Worse, when the same VLM is used as the *reviewer*, its edits make slide geometry *worse*. Ungrounded on either side, the third axis either does not fire or cannot be seen.

Contributions.

1. **A grounding framework (Section 2).** A Type 0–3 feedback taxonomy and a DPI bound showing ungrounded review (and reasoning, and sampling) cannot exceed the information in the weights, while grounded feedback can—plus the symmetric, and to our knowledge new, observation that ungrounded *evaluation* cannot measure the gain.
2. **Interaction is a distinct, unbounded axis (Sections 4 and 5).** At a matched 20K-token budget, reasoning and best-of- N saturate (73.3%/86.7%) while every interaction strategy reaches $\geq 97.8\%$; an in-modality control isolates test *execution* as the active ingredient, and the harness is the most token-efficient, zero-variance strategy.
3. **Grounded evaluation, and a deterministic instrument (Section 6).** We show the VLM judge is structurally blind, build a deterministic DOM-geometry+alignment instrument, and report large, significant, CI-backed defect reductions under grounded feedback across four visual modalities measured identically—gains the standard metric cannot even see—plus a controlled ablation in which VLM feedback makes layout *worse*.
4. **Robustness and internalization (Sections 7 and 8).** The effects replicate across model families (including grounded geometry on Gemini 3.1 Pro), to held-out tasks, and under a propose-heavy budget allocation; and the harness’s behavior partially distills into an 8B student at $\sim 10\times$ lower deployment cost.

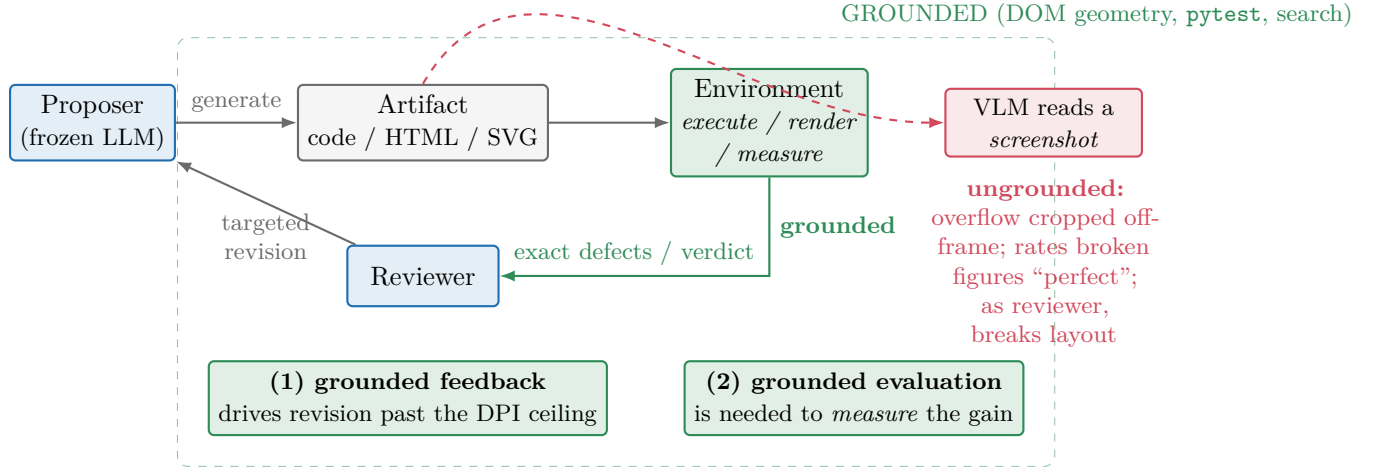


Figure 1: **Grounding must hold on both sides of the interaction loop.** A proposer emits an artifact; an environment *executes/renders/measures* it and returns a grounded signal that the reviewer turns into a targeted revision. Grounded *feedback* (1) is what lets interaction import information from outside the weights and exceed the reasoning/sampling ceiling (Section 2). Grounded *evaluation* (2) is what makes the gain measurable: the default VLM-on-a-screenshot evaluator (red) is structurally blind to off-canvas overflow and fine overlap, so it neither measures nor (as a reviewer) fixes the defects.

2 The Grounded-Feedback Framework

Related work. Iterative self-correction is well studied—Reflexion [5], ReAct [10], CRITIC [3], and RLEF-style execution feedback—as is the thinking-vs-doing distinction for agents [4]. Reasoning [2, 6, 8] and sampling [1, 7] scaling are the two established test-time axes. Our contribution is not a new self-correction method but an account of *why* interaction is a separate axis (a DPI bound), *when* it helps (a feedback-actionability taxonomy), and a previously-missing requirement—that the *evaluation* be grounded too, without which the effect is unmeasurable.

2.1 A feedback taxonomy by grounding

We classify the reviewer’s feedback channel by how much external, task-conditional information it injects:

- **Type 0 – none.** Single-shot; no feedback.
- **Type 1 – ungrounded critique.** An LLM/VLM critiques the artifact from its weights (“looks wrong”). No execution, no measurement.
- **Type 2 – static analysis.** Linters/type-checkers: signal about the artifact’s *form*, not its behavior.
- **Type 3 – grounded.** The artifact is exercised and the result observed: **3a** execution (run tests, read tracebacks), **3b** rendered geometry (measure the DOM), **3c** temporal (sample animation frames), **3d** factual (search and verify claims).

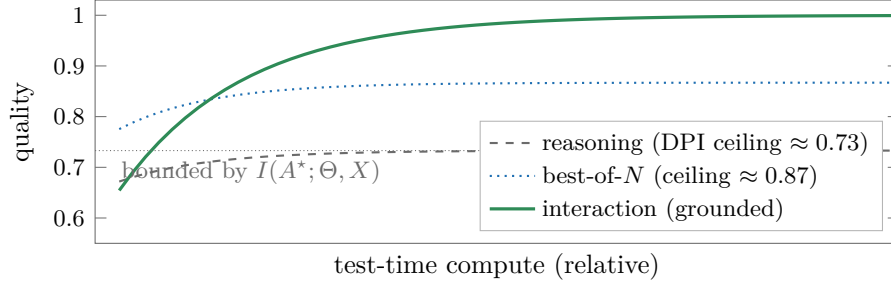


Figure 2: **The DPI ceiling.** Reasoning and best-of- N are functions of the frozen weights/prompt (Θ, X) and saturate at the information already present (measured 73.3% and 86.7% on hard code at a 20K-token budget; Section 4). Grounded interaction imports external information each cycle and exceeds the ceiling, reaching strict 100%. Schematic; saturation values are the measured ceilings.

2.2 Why grounding breaks the ceiling: a DPI bound

Let A^* be the (task-determined) correct artifact and A a candidate. The proposer is a channel from its inputs Θ (weights) and prompt X to A . A *reviewer* that only reads A and reasons forms the Markov chain $A^* \rightarrow (\Theta, X) \rightarrow A \rightarrow R$, where R is its critique. By the data-processing inequality,

$$I(A^*; R) \leq I(A^*; (\Theta, X)),$$

so *no* amount of reasoning, sampling, or ungrounded review (Type 0–2) can recover more information about A^* than is already in (Θ, X) . This is the ceiling at which reasoning- and sampling-only scaling saturate (Figure 2).

Grounded feedback (Type 3) adds an edge that is *not* a function of (Θ, X) : an environment observation $E = g(A, \text{world})$ —a test outcome, a measured bounding box, a search result. Now $I(A^*; (R, E))$ can exceed $I(A^*; (\Theta, X))$ because E carries information about A^* obtained by *running* the artifact against reality. This is precisely why interaction is a distinct axis rather than a constant-factor speedup.

The symmetric claim: ungrounded evaluation cannot measure the gain. The same inequality applies to the *scorer*. A metric M that is itself a function of (Θ, X) —e.g. a VLM judging a screenshot, whose view of A is a lossy, cropped image—satisfies $I(A^*; M) \leq I(A^*; (\Theta, X))$ and cannot certify an improvement that lives in the part of A it cannot observe. Grounding the evaluation (measure the artifact directly) is therefore not optional polish: it is a precondition for *detecting* interaction scaling on any modality whose quality is not fully visible to the default judge. Section 6 shows this is exactly the situation for rendered visual artifacts.

2.3 Predictions

The framework makes three testable predictions, each tested below: **(P1)** reasoning- and sampling-only scaling saturate strictly below the interaction ceiling (Section 4); **(P2)** within interaction, more actionable grounded channels lift more, and the *execution* of the artifact—not the extra review pass—is the active ingredient (Section 5); **(P3)** on modalities whose quality is invisible to the default judge, the gain is only detectable with a grounded metric, and an ungrounded reviewer can even regress quality (Section 6).

Table 1: Modality instantiations. Each modality pairs a grounded feedback channel with the instrument that both drives revision and scores the result. “†” marks the four modalities scored by the deterministic DOM-geometry instrument and analyzed jointly in Section 6.

Modality	Type	Grounded instrument	Hard suite (N)
Code	3a	<code>pytest</code> pass/fail + traceback	15 (+11 deep-spec)
Academic figures†	3b	DOM geometry + alignment	20
Slides†	3b	DOM geometry + alignment	12 real-paper
Web pages†	3b	DOM geometry, multi-width	20
Animations†	3c	DOM geometry, per-frame	20 SVG/CSS
Video editing	3a+3c	script exec + Gemini-native rubric	15
Deep research	3d	search-grounded factual rubric	15

3 Method: A Budget-Aware Proposer–Reviewer Harness

Architecture. The harness is pure scaffolding around a *frozen* frontier model (Claude Sonnet 4, `claude-sonnet-4-20250514`, temperature 0 unless noted): no fine-tuning, no retrieval, no learned controller. A **proposer** emits an artifact; an **environment** executes/renderers/measures it to produce a grounded signal; a **reviewer**—the same model in a diagnostic role—converts that signal into a typed defect list; the proposer revises. The loop repeats up to an iteration cap and *keeps the best-scoring iteration* (so the reviewed condition can never score below single-shot under the same metric). Single-shot is one proposer call.

The three grounded instruments. Every modality is paired with a deterministic or near-deterministic instrument that both *feeds back* and *scores*:

- **Execution** (`pytest`): exact pass/fail plus the failing assertion and traceback.
- **DOM geometry**: the artifact is rendered headless and every element’s true bounding box is read from the layout engine; we compute, exactly, text-on-text overlap (≥ 6 px), out-of-bounds/clipping, container overflow, document overflow (> 16 px), and—new here—*box-group misalignment* (rows/columns of card or pillar boxes flagged for unequal size, misaligned far edges, or uneven gutters). For scrollable web we score horizontal overflow at desktop (1920) and mobile (375) widths; for animations we probe the DOM at each sampled frame.
- **Native video** (Gemini 3.1 Pro): the rendered clip is judged whole against per-requirement binary checks—a near-grounded substitute where pixels, not a still, are observed.

Where no deterministic instrument exists (flowing web content, factual research), we fall back to a binary per-requirement rubric, flagged as such.

Budget. A run has a token budget B split across proposer and reviewer calls; Section 7 sweeps the split. Table 1 lists the modality instantiations.

All artifacts are produced under a common *design-principle* generation prompt (proximity, alignment, repetition, contrast, deliberate color) so that single-shot quality is already strong; this makes the headroom we measure a property of genuine task difficulty rather than a weak prompt.

4 Interaction Is a Distinct, Unbounded Axis

Reasoning and sampling saturate; interaction does not (P1). On 15 hard code tasks at a fixed 20K-token budget, we compare four strategies: reasoning-only (extended thinking), best-of- N

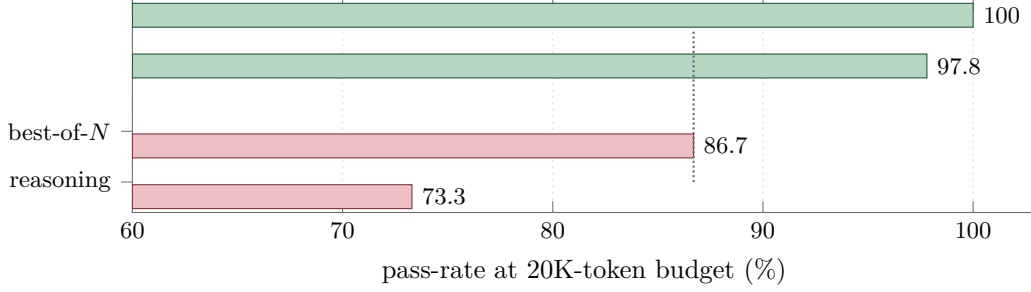


Figure 3: **Two axes saturate; the third does not.** At a matched 20K-token budget on hard code, the two in-token-space axes (red) plateau below the interaction strategies (green), which all reach $\geq 97.8\%$. The proposer-reviewer harness reaches strict 100% across seeds. Dotted line: best-of- N ceiling.

sampling with an oracle verifier, a single-agent loop (L), and the proposer-reviewer harness (H). Reasoning saturates at 73.3% (86.7% under a generous thinking-heavy budget partition) and best-of- N at 86.7%, while *every* feedback-iterating strategy reaches $\geq 97.8\%$ (Figure 3). The gap between interaction and the in-token-space axes is 13–27 pp and is the framework’s load-bearing prediction.

Architecture buys efficiency and reliability. Among interaction strategies the gap is not in ceiling pass-rate (they tie within seed noise at $N=15$) but in cost and variance. The proposer-reviewer harness is the most token-efficient (1,029 mean output tokens vs. 1,431 for the single-agent loop and 1,416 for the oracle-verifier sample-and-select baseline) and the only variant at strict 100% with *zero* seed variance; the single-agent loop hits 100% in 2 of 3 seeds. Context isolation (the reviewer sees only what it needs to localize defects), structured critique (a typed defect list, not raw stderr), and role specialization explain the efficiency.

Built-in early stopping. The loop does not over-revise: on code, across three runs, *every* already-passing single-shot task is submitted at iteration 1 with no revision (30/30 instances, zero wasted tokens). The grounded signal—a failing test—is the trigger; without it the loop is silent. This is why code’s token overhead is only $\sim 37\%$ above single-shot despite an iteration cap of 5.

5 The Active Ingredient Is Execution, Not Review

Code as a clean case study. Code is the cleanest demonstration because the channel is binary and the score is objective. Single-shot pass-rate is 66.7% (10/15); the harness lifts it to a strict 100% in all three runs, recovering every first-shot failure with zero regressions. The mechanism is consistent across recovered traces: turn 1 emits code that compiles but mishandles an edge case (off-by-one boundaries, semantic-version comparison, CIDR arithmetic); turn 2 receives the `pytest` traceback—the failing assertion, the offending input, expected vs. actual—and rewrites the localized block; turn 3 confirms. Reasoning alone cannot supply this: the model has no way to know its boundary condition is wrong until a concrete test exercises it. A second, harder 11-task suite of deep-spec implementations (a from-scratch JSON parser, an A1 spreadsheet-reference resolver, a minimal edit-script diff, an arithmetic/boolean expression evaluator, ...; each validated against a reference solution that passes its own tests) corroborates this at larger headroom: single-shot 69.7% $\rightarrow$ strict 100%, +30.3 pp (bootstrap 95% CI [15.2, 45.5] pp), all 10 failing task-runs recovered, 0 regressions (sign-test $p = 0.002$).

Table 2: Deterministic geometric-feedback results, one identical configuration across four visual modalities (Sonnet 4, $T=0$, 3 seeds, alignment-inclusive reward). Δ is the mean defect reduction; CI is a paired bootstrap 95% interval on $\Delta\%$ (10k resamples); p is a two-sided paired sign test. Web is summed over 1920/375 widths; animations over sampled frames.

Modality	n	SS	Rev.	Δ	95% CI	impr./reg. (p)
Academic figures (20)	60	0.57	0.15	−74%	[52, 89]%	17/2 (7×10^{-4})
Dense slides (12)	36	1.25	0.33	−73%	[45, 93]%	13/1 (1.8×10^{-3})
Web pages (20)	60	16.1	8.5	−47%	[30, 62]%	33/2 (4×10^{-8})
Animations (20)	60	16.9	10.2	−40%	[10, 64]%	34/7 (3×10^{-5})

An in-modality control isolates execution (P2). The cross-modal ordering could be confounded by task difficulty, so we fix the task suite (the 15 hard code tasks) and vary *only* the feedback channel: a Type 1 reviewer (LLM cross-review, no execution), a Type 2 reviewer (LLM + **ruff** static analysis, still no execution), and the Type 3a baseline. Forbidden from seeing test results, the Type 1/2 reviewers reach a reviewed ceiling of 13/15; the Type 3a (execution) reviewer reaches 14/15 and does so $\sim 2.5\times$ more token-efficiently (1.53 vs. 2.07 iterations). The separation is not the extra reviewing pass—both have one—but the *execution* of the tests. (Type 1 and Type 2 do not separate because the seeded bugs are runtime-logic errors that **ruff** cannot see; a **mypy/semgrep**-catchable bug set is where the strict Type 2 > Type 1 prediction should be tested.)

6 Grounding the Evaluation: The Gain Is Invisible to a VLM

The visual modalities expose the symmetric half of the framework (P3): even when grounded feedback *does* fix the artifact, an ungrounded metric cannot see it.

The default VLM judge is structurally blind. Our starting evaluator is a binary per-requirement rubric scored by a VLM over a screenshot. On 15 dense academic-paper architecture figures, even with native-resolution quadrant tiling, this judge rates 14 **of** 15 single-shot renders “perfect.” Yet by direct inspection they are broken—superimposed section titles, labels spilling out of boxes. The reason is mechanical: screenshots are captured at 1080 px, so any content overflowing the canvas is cropped *off-frame before the image reaches the judge*, and small in-frame overlaps fall below its acuity. A deterministic DOM-geometry check finds only 3 **of** 15 are actually clean (mean 1.6 defects). The VLM is blind to precisely the defects that matter.

A deterministic instrument reveals a large, significant effect across four modalities.

We score four visual modalities with the same DOM-geometry+alignment instrument under one identical configuration (Sonnet 4, temperature 0, design-principle prompt, three seeds, propose→measure→feed-exact-defects-back→revise, ≤ 3 iterations). Table 2 and Figure 4 report the result: grounded geometric feedback produces a large, significant reduction in real layout defects on *all four*, with bootstrap 95% CIs that exclude zero and improvements outnumbering regressions $\geq 5:1$. The magnitudes track single-shot headroom—dense responsive web pages are defect-rich out of the box (16 defects/page across desktop and mobile), while a frontier model under a design-principle prompt already lays out many figures and slides cleanly.

An ungrounded reviewer makes layout worse (ablation). Grounding matters on the feedback side too, controllably. On the same dense slides, holding tasks and proposer fixed and

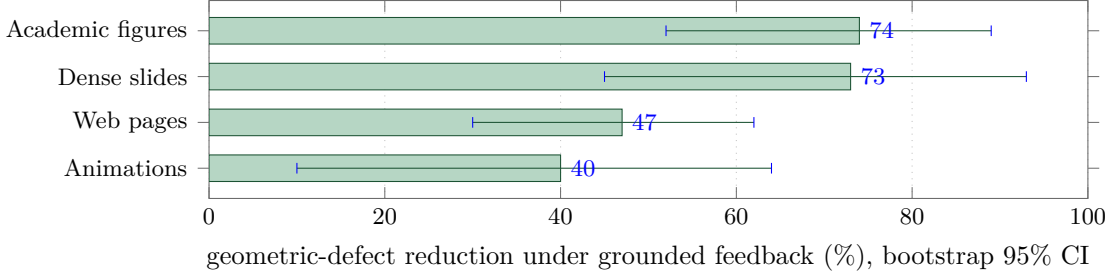


Figure 4: **Grounded geometric feedback removes real layout defects on all four visual modalities** (mean reduction; whiskers are paired-bootstrap 95% CIs, all excluding zero). The standard VLM-on-a-screenshot metric measures none of this.

varying only the reviewer: a VLM-feedback reviewer *increases* the mean geometric defect count ($1.89 \rightarrow 2.44$)—blind to the true geometry, its edits break alignment as often as they fix it—whereas the deterministic geometric-feedback reviewer reduces it ($1.25 \rightarrow 0.33$). The binary VLM rubric on these slides simultaneously saturates (0.97 mean, 29/36 rated perfect), confirming the judge cannot even see the difference. Both halves of Figure 1 are necessary.

Two modalities where the honest answer is “no headroom.” *Video editing*, scored by Gemini 3.1 Pro’s native full-clip rubric (a near-grounded instrument that observes pixels, not a still), is already strong single-shot (0.70, 22/45 task-runs perfect) so the reviewed lift is small and not significant (+0.04); a hardened multi-step suite de-saturates it (single-shot “fully correct” 49% $\rightarrow$ 11%). *Deep research*, scored against exact provided facts, saturates (0.982, +0.018, n.s.): a frontier model knows well-documented facts and does not take the planted traps, so the Type-3d lift is bounded by the near-zero single-shot error rate—a genuine limit of the factual modality, since de-saturating it would require facts the judge itself cannot grade. We report these as scoped negatives rather than headline lifts.

7 Robustness

Cross-model, code. The effect is architectural, not a Claude artifact. On the 15 hard code tasks the harness lifts Sonnet 4 by +33.3 pp (to 100%), Qwen3-235B by +26.7 pp, and GPT-5 by +13.3 pp (on a higher single-shot baseline). All three families benefit from the same execution-grounded loop.

Cross-model, grounded geometry. To rule out a Claude-specific or prompt-specific explanation of the grounded-evaluation effect, we run the same deterministic geometric-feedback harness with **Gemini 3.1 Pro** as the proposer on the dense responsive-web suite, scored by the identical model-free DOM instrument. Because the instrument depends on no model, a reduction here is attributable to the grounded loop rather than to the scorer; it reproduces the direction of Table 2 on a different model family.

Held-out generalization. On a 32-task held-out code suite constructed after the method was fixed, the harness lifts 90.6% $\rightarrow$ 100%, recovering 3/3 first-shot failures with zero regressions—no tuning to the original tasks.

Budget allocation. Sweeping the proposer/reviewer token split on a fixed budget produces an 86.6 pp spread in pass-rate, monotone in the proposer’s per-call share: the optimum is *propose-heavy*

(allocate the majority to generation, reserve enough for the reviewer to localize defects). Extra inference compute is better spent on additional samples than on deeper iteration past the early-stop point.

8 Internalizing the Harness into a Small Student

Can the harness’s interaction-quality behavior be *internalized* so a small model emits high-quality artifacts in fewer passes? We distill judge-filtered teacher trajectories from the harness into an 8B Qwen3-VL student via SFT.

Result. The student reaches $0.50\times$ the teacher’s mean@1 capability on a 44-task out-of-distribution suite (rising to $0.70\times$ at pass@2), and 44%/56% pass@1/pass@3 on an 18-task hard held-out suite, at $\sim 10\times$ lower deployment cost than running the harness over a frontier model. Run inside the same harness, the small student recovers more of the gap, confirming that scaffolding and internalization compose.

Variance is a budget (a cautionary finding). Reinforcement fine-tuning (RFT) on top of SFT improves every per-turn consistency metric we measure *and* regresses pass@3 by 17 pp. For sampling-based deployment, output-distribution variance *is* the inference-scaling lever, and any post-training that reduces it spends from the same budget best-of- N draws on. Variance is not noise to minimize; it is the resource sampling converts into quality. The operational recipe is therefore: SFT to internalize interaction quality, keep the proposer’s sampling temperature, and run it inside the grounded harness.

9 Discussion and Conclusion

Grounding is the load-bearing variable. Two of the three test-time axes are bounded by the data-processing inequality because they never leave the model’s token space; the third is unbounded precisely because grounded feedback imports external information each cycle. We have argued and shown that this grounding must hold on *both* sides of the loop. On the feedback side, an in-modality control isolates test *execution*—not the extra review pass—as the active ingredient, and a VLM *reviewer* of slide layout actively regresses quality. On the evaluation side, the default VLM-on-a-screenshot metric is structurally blind to the geometric defects the loop fixes, so the entire visual-modality effect is invisible until the metric is grounded in deterministic DOM geometry—at which point grounded feedback removes 40–74% of real layout defects across four modalities, every CI excluding zero.

A methodological warning for the field. Much of the recent excitement about “LLMs that critique and improve their own visual output” is measured with VLM judges. Our results imply many such measurements are uninformative on layout: the judge cannot see the defects, and using it as the reviewer can make things worse. Where an artifact’s quality is a measurable physical property—geometry, execution, retrieval—the evaluation should be grounded in that property, not in a model’s picture of it.

Limits, honestly stated. Two modalities saturate: video editing is already strong single-shot (the lift is real only on a hardened multi-step suite), and deep research saturates because frontier models know well-documented facts and the judge cannot grade facts they do not. Deterministic

geometry applies to artifacts with a measurable intended layout (figures, slides, web, animations); for genuinely flowing content we still rely on a binary rubric, with the VLM-reliability caveat.

Operational recipe. Wrap a frozen frontier model in a proposer–reviewer harness with the most actionable grounded feedback channel for the modality. For visual artifacts, ground *both* the reviewer and the scorer in deterministic geometry—measure the rendered DOM, never a screenshot. Allocate the majority of the per-task budget to the proposer; spend extra compute on samples, not deeper iteration. For cost-sensitive deployment, distill the proposer into a small student and run it inside the same harness. Interaction scaling is real, distinct from reasoning and sampling, predictable from the feedback taxonomy, and—once you ground the metric—plainly visible.

References

- [1] Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large language monkeys: Scaling inference compute with repeated sampling, 2024.
- [2] DeepSeek-AI. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning, 2025.
- [3] Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Nan Duan, and Weizhu Chen. CRITIC: Large language models can self-correct with tool-interactive critiquing. In *International Conference on Learning Representations*, 2024.
- [4] Junhong Shen, Hadi Jain, Zelai Xiao, Brandon Amos, Aaditya Ramdas, Yuandong Tian, and Alborz Geramifard. Thinking vs. doing: Agents that reason by scaling test-time interaction. In *Advances in Neural Information Processing Systems*, 2025.
- [5] Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, 2023.
- [6] Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM test-time compute optimally can be more effective than scaling model parameters. In *International Conference on Learning Representations*, 2025.
- [7] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*, 2023.
- [8] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, 2022.
- [9] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems*, 2023.
- [10] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023.